SOFTWARE ENGINEERING

Lab 1: Requirements Engineering & UML Use-Case Modelling

Requirements Table

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PROBLEM STATEMENT #61

Sustainability & Green Tech

Household Carbon Footprint & Target Tracker

Actors: Resident, Sustainability Coach

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest household monthly electricity (kWh) and transport (km) logs to compute total monthly carbon equivalent (CO2e) emissions.	High	Pass: Calculated emissions match EPA conversion factor formulas. Fail: Zero or negative emission values generated.	Core functionality — an accurate baseline CO2e figure is the foundation every other feature (targets, trends, coaching) depends on.
FR-002	Functional	The system shall allow a Resident to log monthly natural gas consumption (in therms or cubic meters) in addition to electricity and transport data.	High	Pass: Gas value is saved and included in that month's CO2e calculation. Fail: Submitted gas reading is missing from the monthly total.	Home heating/gas use is a major residential emission source; omitting it would understate a household's true footprint.
FR-003	Functional	The system shall allow a Resident to set a personal annual carbon-reduction target (in kg CO2e) and shall recompute progress toward that target after every monthly log entry.	Medium	Pass: Progress percentage updates within the same session immediately after a new log is submitted. Fail: Progress value remains unchanged after new data is submitted.	Turns raw emissions data into an actionable, motivating goal, central to the product's purpose as a "target tracker."
FR-004	Functional	The system shall allow a Sustainability Coach to view anonymized, aggregated emission trends across all Resident households assigned to them.	Medium	Pass: Coach dashboard shows per-category average emissions for assigned households without exposing a single resident's raw log. Fail: Dashboard exposes identifiable individual data or returns no data.	Supports the Coach's advisory role while preserving Resident data privacy.
FR-005	Functional	The system shall send an automated reminder notification to a Resident if that month's utility data has not been logged within 5 days after month-end.	Low	Pass: Exactly one reminder is sent per missed month, 5 days after month-end. Fail: No reminder is sent once the 5-day threshold has passed.	Missing months break trend charts and target calculations; proactive reminders keep the dataset complete.
NFR-001	Nonfunctional (Performance & Security)	The annual carbon-reduction milestone chart must render interactive historical comparison graphs in under 200 ms, and all data handling must meet baseline security standards.	High	Pass: Benchmark tests confirm sub-200 ms render latency and pass security checks under simulated peak load. Fail: Render latency exceeds 200 ms or a security check fails.	[Type: Performance & Security] Fast, secure dashboards keep Residents engaged and protect sensitive household usage data.
NFR-002	Nonfunctional (Security & Privacy)	The system shall encrypt all household utility consumption data at rest and in transit using industry-standard encryption (AES-256 at rest, TLS 1.2+ in transit).	High	Pass: Security audit confirms data is encrypted both at rest and in transit. Fail: Any utility data is found stored or transmitted unencrypted.	[Type: Security & Privacy] Utility logs can reveal occupancy and lifestyle patterns; encryption protects Resident privacy and builds trust.